

Freight Rate Prediction Challenge

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1. Data Splitting and Validation Approach To ensure the model could generalize well to unseen data, I utilized a standard Hold-out Validation strategy. The historical dataset (train-test.csv) was split into a **80% Training Set** and a **20% Internal Testing Set** using a random split (via train_test_split with a fixed random state for reproducibility).

I chose this 80/20 split because the dataset is sufficiently large (48,000 records). 80% of the data provides ample volume for the XGBoost algorithm to learn complex, non-linear relationships between locations, distance, and rates, while the remaining 20% provides a robust, statistically significant sample to evaluate metrics (MAE and R-Squared) without risking overfitting.

2. December Prediction Chart Below is the generated chart representing the fixed December predictions for the Lexington to Fort Wayne route:

